Scratch Cheat Sheet

Blocks, loops and events at a glance | Ages 6-10

Scratch is built from colourful blocks that snap together. Find the block, drag it in, click it to test. Every block belongs to a colour group.

MOTION (blue)	
move 10 steps	Moves the sprite forward. Bigger number = bigger jump.
turn 15 degrees	Spins the sprite. 90 = a quarter turn, 180 = turn around.
go to x: y:	Jumps to an exact spot. Centre of the stage is x 0, y 0.
glide 1 secs to ...	Slides smoothly to a spot instead of jumping.
if on edge, bounce	Stops sprites running off the stage.

LOOKS (purple)	
say Hello! for 2 seconds	Shows a speech bubble.
switch costume to / next costume	Changes the picture. Quick costume changes make animation.
change size by / set size to %	Makes the sprite grow or shrink.
show / hide	Makes the sprite appear or vanish.

SOUND (pink)	
start sound	Plays a sound and keeps going with the script.
play sound until done	Waits for the sound to finish first.
change volume by	Makes it louder or quieter.

EVENTS (yellow) - every script starts here	
when green flag clicked	Starts the script when you press the flag.
when space key pressed	Starts when a key is pressed. Great for controls.
when this sprite clicked	Starts when someone clicks the sprite.
broadcast message / when I receive	Lets sprites talk to each other.

CONTROL (orange)	
wait 1 seconds	Pauses the script.
repeat 10	Runs the blocks inside 10 times.
forever	Runs the blocks inside again and again.
if ... then / if ... else	Makes a decision. Only runs if the condition is true.
repeat until ...	Keeps going until something becomes true.
stop all	Ends the game.

Scratch Cheat Sheet (continued)

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SENSING (light blue)	
touching ... ?	True when the sprite touches another sprite or the edge.
key space pressed?	True while a key is held down.
ask ... and wait / answer	Asks a question and stores what the player types.
mouse x / mouse y	Where the mouse pointer is right now.

OPERATORS (green)	
+ - * /	Add, subtract, multiply, divide.
pick random 1 to 10	A surprise number. Used for enemies, dice and prizes.
> < =	Compare two things. Used inside if blocks.
and / or / not	Join several conditions together.

VARIABLES (red)	
Make a Variable	A named box that remembers a number or word, like score or lives.
set score to 0	Resets the box. Do this when the green flag is clicked.
change score by 1	Adds to the box, e.g. +1 every time you catch a star.
My Blocks	Build your own block so you don't repeat the same code.

Mini challenge: Catch the Star (15 minutes)

1. Add a basket sprite. Use **when green flag clicked** + **forever** + **set x to mouse x**.
2. Add a star sprite. Make it fall: **forever**, **change y by -5**.
3. **if touching basket** then **change score by 1** and **go to x: (pick random -200 to 200) y: 170**.
4. Bonus: add a Lives variable, a game over message, and a sound when you catch a star.

DEBUGGING: when it doesn't work	
Nothing happens	Did you start with an Events block? Did you click the green flag?
Sprite moves too fast	Add **wait 0.1 seconds** or use a smaller number.
Score never resets	Add **set score to 0** under **when green flag clicked**.
Two scripts fight	Check you haven't got two **forever** loops moving the same sprite.
Stuck?	Click one block at a time to see which one is the problem.

Tip for parents: ask your child to explain what each script does out loud. Explaining code is one of the best ways to learn it. Practise at scratch.mit.edu (free).